

Graded Project - Part 1

AI Project Functional Methodologies

Functional framing of a customer churn prediction project at RetailGenius

A complete functional framing covering strategy and KPIs, data and model design, deployment and monitoring, team organisation, governance and communication, and the project management methodology - including a Trello Kanban simulation of the project from framing to run.

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Case study	Predicting Customer Churn at RetailGenius
Deliverable	Part 1 report - group dedicated repository
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Part 2 of the graded project implements this framing as a production-ready Python project (cookiecutter structure, PEP8 tooling, dependency management, Sphinx documentation and MLflow tracking, registry and serving).

1. Executive summary

RetailGenius is a fast-growing global e-commerce marketplace that has launched a strategic programme to turn its data into value. The first AI project of that programme is a **customer churn prediction** capability: identifying, ahead of time, the customers who are about to stop buying, so that retention actions can be taken while they are still reachable.

This report is the **functional framing** of that project. It answers a single question - what has to be true, organisationally and technically, for a churn model to actually change the retention outcome at RetailGenius - and it does so before a single line of modelling code is written. Part 2 of the graded project implements the resulting design.

Framing decisions taken in this report

Topic	Decision	Why
Churn definition	Behavioural: no order in the 90 days following the scoring date, for a customer active in the previous 12 months	E-commerce is a non-contractual setting - there is no cancellation event, so churn must be defined, not observed
Primary business goal	Reduce voluntary churn of value-generating customers by 15% relative on targeted segments in 12 months	Retention is materially cheaper than acquisition and directly moves GMV and LTV
Primary model metric	PR-AUC, with lift in the top decile as the decision metric	The positive class is a minority and only the top of the ranking is ever actioned
Model family	Gradient boosting on tabular features (LightGBM / XGBoost), logistic regression as baseline	Best accuracy-per-effort on heterogeneous tabular data; SHAP gives the reason codes the business needs
Serving pattern	Weekly batch scoring into the CRM, plus a low-latency REST endpoint for customer support	Retention campaigns are planned weekly; support needs the score at the moment of contact
Success measurement	Randomised control holdout on every campaign	A model that ranks well can still create zero incremental value - only a holdout proves uplift
Methodology	CRISP-DM lifecycle, executed with Scrum for build and Kanban for run, with five phase gates	ML work is iterative and uncertain; gates keep that iteration inside a budget

The trap this framing is designed to avoid

The default failure mode of a first churn project is a technically good model that nobody uses: scores that never reach a campaign tool, a threshold nobody agreed, discounts handed to customers who were never going to leave, and no control group to tell the difference. Every section below is written against that failure mode.

2. Business context and problem definition

2.1 What the interviews and workshops told us

The material gathered before this framing - stakeholder interviews, workshops, and the analysis of the existing data landscape - converges on a small number of facts that constrain the project.

Source	Key input	Consequence for this project
Business stakeholders	Decisions are made on customer demographics, purchase history, product performance and campaign effectiveness; KPIs are conversion, retention, revenue growth and satisfaction	The churn model must plug into an existing KPI frame rather than introduce a parallel one; retention is already a board-level metric
Data team	Sources are user interactions, seller activity, order processing and support; the pain points are silos, data quality, privacy and the cost of integration	Data preparation, not modelling, is the critical path. Budget and sprint capacity must reflect that
Technology team	Mixed SQL / NoSQL, cloud storage and data lakes; ETL with Nifi and Talend; scalability is an active concern	The solution must reuse the existing ETL stack and be designed for batch first; a real-time feature store is out of scope for v1
Workshops	Product wants real-time behavioural data, marketing wants demographics and campaign engagement, the data team insists on governance and quality, infrastructure wants easy integration and security	The feature set must span behaviour, transactions, marketing and service; governance is a hard requirement, not a later phase
Documentation analysis	Data dictionaries, a metadata catalogue, data flow diagrams and a preliminary quality assessment already exist	A real head start: lineage and semantics are documented, so the project can focus on point-in-time correctness rather than on discovery

2.2 Defining churn in a non-contractual business

RetailGenius has no subscription and no cancellation event. A customer who has not ordered for six weeks may be lost, or may simply have a long purchase cycle. Churn therefore has to be **defined as a business rule** and that rule becomes the label the model learns. The definition below was chosen by combining the observed inter-purchase interval distribution with what the retention team can act on.



T - 12 months

scoring date T

T + 90 days

Churn label = 1 if the customer places no order between T + 7 d and T + 90 d, given at least one order in the observation window.

Element	Value	Rationale
Population	Customers with at least one order in the last 12 months	Excludes one-off and dormant accounts that would dominate and dilute the signal
Observation window	12 months ending at the scoring date T	Long enough to capture seasonality and a full purchase cycle for most categories
Operational gap	7 days after T	Features must be frozen before the retention action is decided; the gap prevents the model from learning from events that happen after the decision
Prediction window	90 days (T+7d to T+90d)	Covers roughly the 80th percentile of the inter-purchase interval; short enough for a campaign to still matter
Label	1 = no order placed in the prediction window	Simple, auditable, and computable from order data alone
Segmentation	Separate treatment for new customers (tenure < 90 days)	Early-life churn has different drivers and almost no history; a rule-based or dedicated model is more honest than extrapolating

Definition is a business decision, not a data science one

The 90-day window is a parameter with direct financial consequences: a shorter window inflates the churn rate and floods campaigns with false alarms, a longer one produces warnings that arrive too late to act on. It is validated with marketing and finance at gate G1 and re-tested annually with a sensitivity analysis over 60 / 90 / 120 days.

3. Project strategy

3.1 Strategic objectives

The project has one commercial objective and four enabling ones. They are written so that each can be falsified at the end of the year.

#	Strategic objective	Business rationale	Target at 12 months
O1	Reduce voluntary churn of active customers	Retaining an existing customer is several times cheaper than acquiring a replacement, and retention compounds into LTV	-15% relative churn on targeted segments vs. control
O2	Move retention from reactive win-back to proactive prevention	Today the customer is contacted after they have already gone silent, when intent has decayed	80% of retention budget spent before the churn event
O3	Protect high-value revenue first	Churn is not uniform - a small share of customers carries most of the GMV	+8% retained GMV in the top two value deciles
O4	Make retention spend measurable	Campaigns currently run without a control group, so incremental effect is unknown	100% of retention campaigns run with a randomised holdout
O5	Build reusable data and AI foundations	This is the first project of the strategic data programme; the plumbing should outlive it	Feature pipelines, registry and monitoring reused by at least two further use cases

Explicit non-goals. The project does not aim to maximise model accuracy for its own sake, does not attempt to model **why** a customer churns causally in version 1, and does not automate the retention decision - a human owns the offer policy. Naming non-goals is what keeps the scope, and therefore the cost, under control.

3.2 Key performance indicators

KPIs are split into four layers. The distinction matters: a model can be excellent on layer 2 and still fail on layer 1, which is the only layer the steering committee cares about.

Layer 1 - Business KPIs (the ones that justify the investment)

KPI	Definition	Baseline / target	Cadence
Churn rate (90 d)	Share of active customers with no order in the following 90 days	Baseline measured over the last 4 quarters; -15% relative on targeted segments	Monthly
Incremental retention uplift	Retention rate of the treated group minus the randomised control group	> +3 points, statistically significant	Per campaign
Retained GMV	GMV in the 90 days after contact, treated minus control, extrapolated	+8% in top two value deciles	Quarterly
Campaign ROI	(Incremental margin - incentive cost - operating cost) / total cost	> 1.5 on the top three deciles	Per campaign
Incentive efficiency	Share of discount budget spent on customers who would have stayed anyway	Reduce vs. current untargeted baseline	Quarterly

KPI	Definition	Baseline / target	Cadence
Contact fatigue	Unsubscribe and opt-out rate among contacted customers	No increase vs. current level	Monthly

Layer 2 - Model KPIs

KPI	Why this one	Acceptance threshold
PR-AUC (primary)	Robust under class imbalance, unlike ROC-AUC which flatters a minority positive class	Beats the business rule baseline by $\geq 20\%$ relative
Lift @ top 10%	The operational question: how much richer in churners is the list we actually contact	$\geq 2.5\times$ the base rate
Recall @ top 10%	Share of all future churners captured in the contacted list	$\geq 30\%$
Precision @ k	Drives the cost of wasted incentives	Monitored, traded off against recall by expected value
Calibration (Brier score, reliability curve)	Probabilities are multiplied by margin to prioritise; if they are not calibrated the prioritisation is wrong	Brier improves on the baseline; reliability curve within tolerance
Stability (PSI on scores)	Detects a model whose score distribution has quietly moved	PSI < 0.1 week over week
Slice performance	Recall must not collapse on a segment (country, tenure, device, value decile)	No slice below 60% of the global recall

Layers 3 and 4 - Operational and adoption KPIs

Layer	KPI	Target
Operational	Scoring pipeline SLA - scores available before the Monday campaign build	99% of weekly runs on time
Operational	Score freshness - age of the newest event used in a score	< 48 hours
Operational	REST endpoint p95 latency / availability	< 300 ms / 99.5%
Operational	Coverage - share of the eligible base scored	$> 98\%$
Operational	Time to retrain and redeploy	< 5 working days
Adoption	Share of high-risk customers actually receiving an action	$> 70\%$ within 7 days of scoring
Adoption	Support agents using the score in the console	$> 50\%$ of eligible contacts
Adoption	Business users able to read the reason codes without help	Measured by survey after training

3.3 How AI contributes to customer retention

AI is not the retention strategy; it is what makes a retention strategy affordable. RetailGenius cannot contact everybody, and the cost of a retention offer is real margin. The model converts an undifferentiated base into a ranked, explained and prioritised list.

Value lever	What AI adds	Concrete effect at RetailGenius
Prioritisation	A ranking of risk that beats tenure or recency rules, learned from dozens of weak signals at once	The retention budget is concentrated on the deciles where it changes the outcome
Anticipation	A 90-day forward view instead of a backward-looking silence report	Contact happens while the customer still has intent, not after they have moved to a competitor
Explanation	Per-customer reason codes from SHAP (delivery delays, price sensitivity, category exit, support friction)	Marketing chooses a relevant treatment; product gets a ranked list of churn drivers to fix
Personalisation	Risk score combined with value and preferences to select the offer tier	A high-value price-sensitive customer gets a different action from a low-value lapsed browser
Automation	Scores pushed into CRM journeys and the support console without manual export	Retention becomes a repeatable weekly process rather than a quarterly campaign
Learning loop	Campaign outcomes flow back as training and evaluation data	Each cycle improves both the model and the offer policy
Next step (phase 2)	Uplift / causal modelling to target the persuadable , not merely the at-risk	Stops spending margin on customers who would have stayed, and on 'sleeping dogs' who react badly to being contacted

Risk score is not a retention plan

A churn probability answers 'who' - it does not answer 'what should we do about it'. The functional design therefore pairs every score band with an agreed action (see 4.3), and the decision to act is governed by an offer policy owned by marketing, not by the model.

4. Project design

This section covers the four design axes required by the brief - data, models, deployment and monitoring. The target architecture below is the reference for all four.

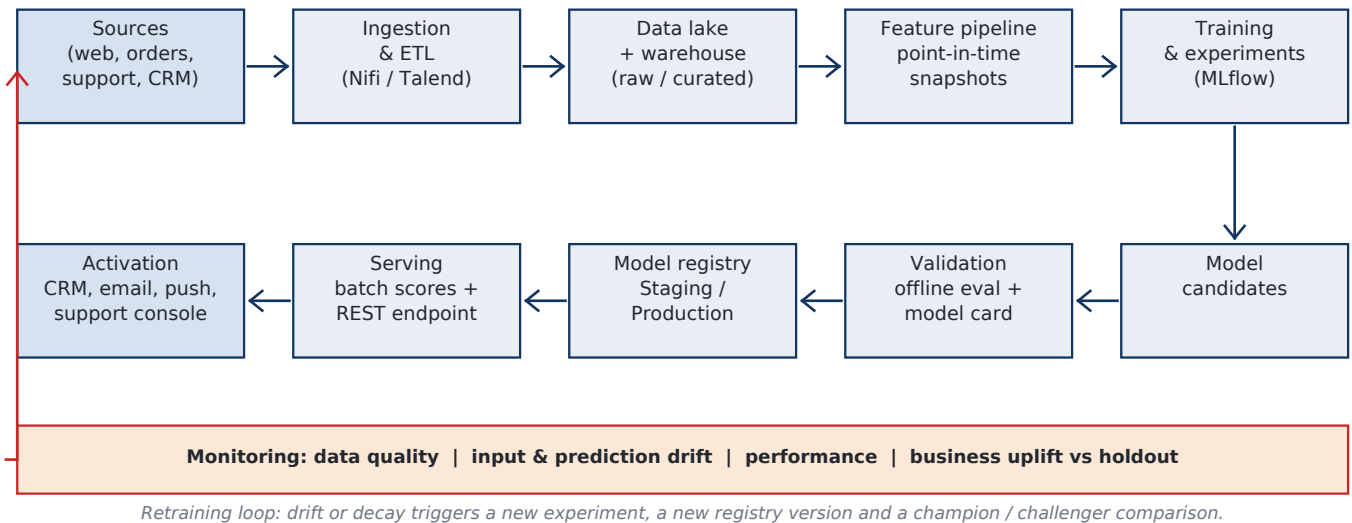


Figure 1 - Target architecture for the churn prediction project. Version 1 is batch-first: the REST endpoint reads pre-computed features rather than computing them online.

4.1 Data

Relevant data sources

Churn in e-commerce is preceded by observable friction. The sources below are ordered by expected predictive value, which is also the order in which they should be integrated - the project does not wait for the full list before training a first model.

Source	Type	Examples of signal	Refresh	Expected value
Orders and transactions	Internal	Recency, frequency, monetary value, basket size, category mix, inter-purchase interval and its recent trend, discount dependence	Daily	Very high
Web and app behaviour	Internal	Sessions, product views, searches, cart additions and abandonments, wishlist activity, session length decay, app uninstall proxies	Daily (batch)	Very high
Service friction	Internal	Support tickets, reopened tickets, resolution time, CSAT, complaints, refund disputes	Daily	High
Delivery and fulfilment	Internal	Late deliveries, failed deliveries, return rate, cancelled orders, out-of-stock exposure	Daily	High
Marketing engagement	Internal	Email opens and clicks, push opt-out, campaign exposure, response history, contact frequency	Daily	Medium-high
Customer profile	Internal	Tenure, acquisition channel, country, loyalty tier, consent and preferences, device mix	Weekly	Medium
Payment and pricing	Internal	Payment failures, price paid vs. reference price, sensitivity to promotions	Daily	Medium

Source	Type	Examples of signal	Refresh	Expected value
Seller and product quality	Internal	Rating of sellers bought from, defect and return rate of purchased items, review scores	Weekly	Medium
Reviews and sentiment	External	Sentiment on third-party review platforms, brand mentions	Weekly	Medium (v2)
Competitor and market data	External	Competitor price index on comparable baskets, market trend reports	Weekly	Medium (v2)
Economic and demographic	External	Purchasing power and demographic indicators by area, from open data	Quarterly	Low-medium
Social and weather	External	Social engagement, weather for weather-sensitive categories	Varies	Low - v3, evaluate before paying for it

Feature families derived from these sources

- **RFM and its derivatives** - recency, frequency, monetary value, but above all their **trend**: the ratio of activity in the last 30 days to the previous 90 is usually stronger than any level.
- **Engagement decay** - slope of sessions, views and opens over the observation window.
- **Breadth** - number of distinct categories and sellers bought from; narrowing breadth often precedes exit.
- **Friction counters** - late deliveries, returns, tickets, payment failures, weighted by recency.
- **Price and promotion behaviour** - share of orders placed with a discount, reaction to price changes.
- **Lifecycle** - tenure, days since first and second order, position in the expected purchase cycle.
- **Marketing pressure** - contacts received per month and the response to them, to avoid confusing fatigue with disengagement.

Data challenges and mitigations

Challenge	Why it hurts this project	Mitigation
Data silos and identity resolution	Orders, clickstream and support use different customer keys; a customer seen as three identities has three broken histories	Build a customer ID graph in the curated layer first; measure the match rate and publish it as a data quality KPI
Target leakage	The single most dangerous failure. Any feature computed after T - cancellation flags, 'last order date' recomputed today, win-back campaign exposure - produces a model with excellent offline scores and no real skill	Point-in-time feature snapshots, a strict cutoff at T for every join, a written leakage checklist reviewed at gate G3, and scepticism towards any metric that looks too good
Label definition ambiguity	Churn is a business rule; a badly chosen window trains the model on the wrong question	Agreed at G1 with marketing and finance, revisited with a 60/90/120-day sensitivity analysis
Class imbalance	Churners are a minority; accuracy is meaningless and a naive model predicts 'nobody churns'	Rank-based metrics (PR-AUC, lift), class weights, threshold chosen on expected value; resampling only if it demonstrably helps on the out-of-time set
Data quality - missing values, outliers, duplicates	Already identified in the preliminary assessment; silently propagates into features	Automated expectations on every pipeline run (schema, ranges, null rates, duplicate keys), run blocked on critical failure, quality dashboard owned by the data steward

Challenge	Why it hurts this project	Mitigation
Training / serving skew	Features computed differently in the training notebook and in the scoring job produce a model that degrades on day one	One shared transformation library used by both paths (enforced in Part 2 by the project structure), plus a nightly comparison of a sample of features
Cold start and thin history	New customers have almost no observation window	Excluded from the main model; handled by a separate early-life rule set with its own KPIs
Seasonality and shocks	Peak season, sales periods and one-off events distort both features and labels	Out-of-time validation across a seasonal boundary, seasonal calendar features, and a documented decision not to retrain on an anomalous period
Privacy and GDPR	Behavioural profiling of identified individuals across the EU; consent scope differs by source	DPIA before production, legitimate-interest assessment, pseudonymised customer keys in the feature store, no special-category data, consent and suppression enforced at activation , documented retention periods, right to object honoured
External data reliability and cost	Scraped or purchased data breaks silently and carries licence conditions	Treat every external source as optional: the pipeline degrades gracefully, and a source is only paid for after an offline value test

Governance is inherited, not invented

RetailGenius already has a governance framework with defined roles and ownership, a metadata catalogue with lineage, and data dictionaries. The project reuses them: every new feature table is registered in the catalogue with an owner, a definition and a lineage entry before it is allowed into a production model. This is the cheapest available protection against the silos the data team described.

4.2 Models

Candidate models

Model	Strengths	Limits	Role here
Business rule (e.g. no order for 60 days)	Free, instantly understood, already exists implicitly	No ranking, no probability, poor precision	Mandatory baseline - the model must beat it or the project has no case
Logistic regression	Transparent coefficients, calibrated by construction, fast, trivial to deploy	Needs manual feature engineering, misses interactions	Reference baseline and sanity check on feature quality
Random forest	Handles non-linearity, robust to outliers, little tuning	Larger models, weaker calibration, less accurate than boosting on this shape of data	Secondary comparison
Gradient boosting (LightGBM / XGBoost)	Best accuracy-per-effort on heterogeneous tabular data, native handling of missing values, fast to retrain, SHAP support	Needs calibration, can overfit without out-of-time validation, less transparent than a linear model	Recommended production model for v1
Survival analysis (Cox, accelerated failure time)	Answers when , not only whether; handles censored customers correctly	Heavier to operate, harder to explain to marketing	Evaluated in v2 for campaign timing

Model	Strengths	Limits	Role here
Sequence models (LSTM / transformer on event streams)	Learn behavioural patterns directly from raw event sequences	Data-hungry, expensive, opaque, requires a mature pipeline	Out of scope for v1; revisit once the data foundation is stable
Uplift / causal models (two-model, X-learner)	Target the persuadable rather than the at-risk; directly optimises campaign margin	Requires randomised campaign data that does not exist yet	Phase 2 - the holdouts introduced now are what make it possible later

The choice of gradient boosting is deliberate and modest. The brief for Part 2 states that the best ML performance is not the goal; the functional argument is the same one - a well-operated, well-explained boosted tree that marketing trusts creates more value than a marginally more accurate model nobody can interpret or retrain.

Training, validation and testing

Step	Protocol
Dataset construction	One row per customer per scoring date, built from point-in-time snapshots. Several scoring dates are stacked to give the model multiple seasonal contexts, with customer-level grouping kept intact
Splitting	Out-of-time split, never random: train up to T-6 months, validate on T-6 to T-3, test on the most recent 3 months, unseen until the end. A random split would leak the future into the past and overstate performance
Cross-validation	Grouped by customer inside the training window only, to prevent the same customer appearing in both folds
Imbalance	Class weights / scale_pos_weight first. Synthetic oversampling is tested but only accepted if it improves the out-of-time set, since it often improves only the resampled validation set
Hyper-parameter search	Time-boxed randomised or Bayesian search with a fixed budget of runs, all tracked in MLflow with parameters, metrics, data version and code commit
Threshold selection	Not 0.5 and not the F1 optimum: the operating point is chosen on an expected value curve using the cost of a contact, the margin of the offer and the value of the customer, then converted into a contactable volume marketing can absorb
Calibration	Isotonic or Platt scaling fitted on the validation window and checked with a reliability curve, because scores are multiplied by customer value to prioritise
Robustness and fairness	Performance reported per slice - country, tenure band, value decile, device, acquisition channel - with an explicit floor per slice. Sensitive attributes are neither used as features nor ignored in evaluation
Explainability	Global SHAP importance for the business review, per-customer SHAP reason codes delivered alongside the score, and a stability check that reason codes do not swing between retrains
Acceptance	A model is accepted at gate G3 only if it beats the business rule baseline on PR-AUC and lift, passes the leakage checklist, meets every slice floor, and is documented in a model card

Model versioning and serving

Reproducibility is a functional requirement, not a developer preference: when marketing asks in March why a customer was contacted in January, the exact model, data and code must be recoverable.

Artefact	Versioned how	Purpose
Code	Git, feature branches, pull requests, protected main, tagged releases	Traceability and collaboration (GitHub, as required in Part 2)
Data	Immutable dated snapshots of the training table, hash recorded as an MLflow tag	The same training set can be rebuilt months later
Experiments	MLflow tracking - parameters, metrics, artefacts, plots, environment, git SHA	Comparison of runs and audit of what was tried
Models	MLflow Model Registry with stages None -> Staging -> Production -> Archived, and a semantic version	One authoritative answer to 'what is in production'; rollback is a stage transition, not a redeployment
Environment	Pinned dependency file (conda / poetry) and a container image per release	Identical behaviour in training and serving
Documentation	Model card per registered version: purpose, data, metrics per slice, limitations, owner, review date	The artefact the model validation committee actually reviews

Serving. Two complementary paths share one model version. The weekly batch job scores the eligible base and writes scores, bands and reason codes into the CRM and the analytics warehouse. A REST endpoint - the MLflow inference server in Part 2, containerised for production - serves the same model to the support console for on-demand scoring. Both read features from the same curated tables, which is what keeps them consistent.

4.3 Deployment

Deployment strategies considered

Strategy	How it works	Fit for this project
Batch scoring	Scheduled job scores the whole eligible base and writes to the CRM	Chosen for v1. Retention campaigns are planned weekly; batch is cheap, simple to monitor and easy to rerun
Real-time REST endpoint	Model served behind an API, called per customer on demand	Chosen as a complement for the support console and for on-site triggers; v1 uses pre-computed features to avoid an online feature store
Streaming / event-triggered	Score recomputed on a behavioural trigger (e.g. a complaint)	Deferred to v2 - it requires online features and a much heavier operational commitment
Embedded in the CRM tool	Model logic exported into the campaign tool	Rejected - it fragments versioning and makes monitoring impossible
Shadow deployment	New model scores in parallel without driving any action	Used before every promotion: one full cycle in shadow, comparing distributions and agreement with the champion
Canary / progressive rollout	New version drives a small share of the base, then more	Used for the first production release and for any major feature change
Blue / green	Two environments, instant switch and instant rollback	Used for the API layer, where availability matters
Champion / challenger	Challenger scored offline on the same population every cycle	Permanent practice - it is what makes retraining a routine rather than an event
A/B test with holdout	A random share of high-risk customers is deliberately not contacted	Mandatory. Without it, uplift cannot be measured and O4 fails by construction

From score to action

Risk band	Population	Action owner	Action
High risk / high value	Top decile, top two value deciles	Retention manager	Personalised offer or human contact, with a 10% randomised holdout
High risk / standard value	Top decile, remaining value deciles	CRM automation	Automated journey: reminder, targeted recommendation, moderate incentive
Medium risk	Deciles 2-3	CRM automation	Low-cost nudge - content, recommendations, no margin given away
Low risk	Remainder	-	No retention action; normal marketing pressure
Any risk, inbound contact	Customer calls support	Support agent	Score band and top three reason codes shown in the console, with a retention gesture allowed within policy

Production considerations

- **Consent and suppression enforced at activation**, not in the model: a customer may be scored for analytics yet be non-contactable. The activation layer applies consent, frequency caps and suppression lists.
- **CI/CD and infrastructure as code** - tests, linting and packaging on every commit; environments reproducible from code.
- **Feature parity between batch and API** enforced by a shared transformation library and a daily sample comparison.
- **Scalability** - the eligible base grows; the scoring job is partitioned and its runtime is tracked as an operational KPI, in line with the technology team's concern about data volume.
- **Security** - pseudonymised keys, encryption at rest and in transit, least-privilege access to the feature store, secrets in a vault, full audit trail of who scored what.
- **Rollback path** - previous registry version promoted back, batch re-run from the last good snapshot; tested before go-live, not during the incident.
- **Human in the loop** for high-value customers, where an inappropriate automated offer costs more than the model saves.
- **Cost control** - compute cost per scoring run and per experiment tracked as an MLflow tag and reviewed monthly.
- **Run handover** - runbook, alert routing, on-call rotation and documentation (generated with Sphinx in Part 2) delivered as part of the definition of done, not after go-live.

4.4 Monitoring

A churn model degrades quietly. Its inputs move with the catalogue, the marketing calendar and the market, and its ground truth arrives 90 days late - so monitoring cannot rely on accuracy alone.

Layer	What is watched	Frequency	Alert threshold	Owner
Infrastructure	Job success, runtime, API latency and error rate	Every run	Any failure; runtime +50% vs. median; p95 > 300 ms	MLOps engineer
Data quality	Schema, null rates, duplicates, volume, freshness, ID match rate	Every run	Critical check fails -> run blocked and alert raised	Data engineer / steward
Input drift	PSI and KS per feature, on the top 30 features by importance	Weekly	PSI > 0.1 investigate, > 0.2 retrain candidate	Data scientist
Prediction drift	Score distribution, mean score, share above the operating threshold	Weekly	PSI > 0.1 or mean score shift > 2 sigma	Data scientist

Layer	What is watched	Frequency	Alert threshold	Owner
Performance (delayed)	PR-AUC, lift @10%, recall @10% on cohorts whose 90-day window has closed	Monthly, per closed cohort	Lift below 2.0 or -20% vs. validation	Data scientist
Early proxies	Short-horizon signals (30-day inactivity among predicted churners) while the real label matures	Weekly	Divergence from the historical proxy relationship	Data scientist
Business impact	Uplift treated vs. control, campaign ROI, incentive efficiency	Per campaign + quarterly	Uplift not significant two campaigns in a row	Retention manager
Fairness / slices	Recall and lift per country, tenure, value decile, device	Monthly	Any slice below 60% of global recall	Data scientist
Adoption	Share of high-risk customers actioned, agent usage of the console	Monthly	Below target two months running	Product owner

Handling model drift

Drift is not a single phenomenon and the response depends on which kind it is. The team diagnoses before it retrains - retraining on corrupted data is the most common way to make a degraded model worse.

Observed	Likely cause	Response
Sudden break in one feature	Pipeline or upstream schema change, tracking bug	Fix the pipeline; do not retrain. Rescore affected cohorts once corrected
Gradual input drift, stable performance	Population evolution	Log, watch, include in the next scheduled retraining
Input drift with performance decay	Genuine concept drift - behaviour has changed	Trigger an off-cycle retraining on a recent window, challenge the champion, promote if it wins
Seasonal swing	Peak period, sales event	Compare against the same period last year rather than last month; avoid retraining on an anomalous window
Performance decay without input drift	Label or definition change, campaign cannibalisation, the model's own effect on behaviour	Re-examine the label rule and the campaign log; the holdout is the reference, not the treated population

Retraining and maintenance policy

- **Scheduled retraining** every month on a rolling window, fully automated and tracked in MLflow.
- **Triggered retraining** when PSI > 0.2 on important features or lift @10% falls below 2.0.
- **Champion / challenger** evaluation every cycle; promotion only if the challenger wins on the out-of-time set and passes the slice floors.
- **Shadow period** of one full cycle before any promotion to Production.
- **Annual re-framing** - the churn definition, the feature set and the segmentation are re-examined once a year with the business, since the business itself changes.
- **Contamination guard** - the treated population is biased by the campaigns themselves, so the control holdout is the reference population for evaluation and, where possible, for label construction.
- **Documented decommissioning** - if the model stops creating measurable uplift for two quarters, the project is stopped or re-framed rather than maintained out of habit.

5. Project team

The team is deliberately small and cross-functional: one squad with a single backlog, not a sequence of handovers between a data team, an IT team and a marketing team. The interviews showed that RetailGenius already has the individual competences; what the project adds is a shared objective and a shared cadence.

5.1 Roles, expertise and effort

Role	Responsibility on this project	Key skills	Effort
Product Owner (business)	Owns the value case, the churn definition, the backlog priority and the offer policy; represents marketing and retention	E-commerce retention, CRM, prioritisation, ability to say no	0.5 FTE
AI Project Manager / Scrum Master	Runs the cadence, the gates, the risk register, budget and planning; removes impediments	Agile facilitation, ML project literacy, stakeholder management	0.5 FTE
Lead Data Scientist	Owns the modelling approach, validation protocol, leakage checklist and model card	Statistics, ML, experimental design, causal thinking, Python	1 FTE
Data Scientist	Feature engineering, experiments, explainability, slice evaluation	Python, scikit-learn, LightGBM, SHAP, MLflow	1 FTE
Data Engineer (x2)	Ingestion, curated layer, identity resolution, point-in-time feature pipelines, data quality checks	SQL, Spark, Nifi / Talend, orchestration, data modelling	2 FTE
ML / MLOps Engineer	Packaging, CI/CD, registry, batch and API serving, monitoring, cost and performance	Docker, CI/CD, MLflow, cloud, observability, Python engineering	1 FTE
Data Analyst	Baseline churn measurement, campaign and holdout analysis, business dashboards	SQL, BI tooling, statistics, A/B testing	0.5 FTE
Data Steward / Governance	Definitions, catalogue entries, ownership, quality KPIs, compliance with the existing framework	Data governance, metadata, domain knowledge	0.3 FTE
DPO / Legal (advisory)	DPIA, legal basis, retention and consent rules, review of the activation logic	GDPR, AI regulation, risk assessment	0.2 FTE
Security / IT (advisory)	Access control, secrets, network and infrastructure review	Cloud security, IAM	0.2 FTE
Retention / CRM Manager	Designs the campaigns, owns the incentive budget and the holdout discipline	CRM tooling, campaign design, offer economics	0.5 FTE
UX Designer (part-time)	How the score and reason codes appear in the support console	UX for internal tools, information design	0.2 FTE

Core squad of roughly 8 FTE, with advisory roles called in at defined gates. The two data engineers reflect the fact that data integration, not modelling, is the critical path.

5.2 RACI across the project phases

Activity	PO	AI PM	Data Sci.	Data Eng.	MLOps	CRM	Gov / DPO
Define churn rule and KPIs	A	C	R	I	I	C	C
Data access and integration	I	C	C	A/R	C	I	C
Feature engineering	C	I	A/R	R	I	C	I
Model training and validation	I	I	A/R	C	C	I	I
Privacy and compliance review	C	C	C	C	I	I	A/R
Production deployment	I	C	C	C	A/R	I	C
Campaign design and holdout	A	I	C	I	I	R	C
Monitoring and retraining	I	C	R	C	A	I	I
Value measurement and reporting	A	R	C	I	I	R	I

R = responsible, A = accountable, C = consulted, I = informed.

5.3 Ensuring cross-functional collaboration

- **One team, one backlog** - data engineering, data science and MLOps work from the same board with the same sprint goal, so that the pipeline work that unblocks modelling is prioritised against modelling work, not in a separate queue.
- **An embedded business voice** - the retention manager attends refinement and demo, and is the person who signs off the churn rule and the score bands.
- **A shared vocabulary** - 'active customer', 'churn', 'order' and 'value decile' are defined once in the data dictionary; every KPI on every slide resolves to those definitions.
- **A definition of done that spans functions** - a story is done when the code is reviewed, the feature is registered in the catalogue, the pipeline has quality checks, the run is tracked in MLflow and the documentation is updated.
- **Pairing across boundaries** - data scientist and data engineer pair on feature pipelines; MLOps and data scientist pair on packaging, which is how training / serving skew gets caught early.
- **A written decision log** - every framing decision, its date, its owner and its rationale, so that a question in month six does not restart the debate.

5.4 Keeping the team aligned with the strategy

- Objectives O1-O5 are translated into quarterly OKRs that are visible on the board; every epic declares which objective it serves, and work that serves none is a candidate for removal.
- The sprint goal is written in business terms ('we can rank the top decile by risk on real data'), not in task terms.
- The value case is re-tested at each gate with the latest numbers - if the expected uplift no longer covers the cost, the project stops. Stopping is an acceptable outcome and is easier to accept when it is planned for.
- Model KPIs are always reported next to business KPIs, so nobody optimises PR-AUC in isolation.
- Onboarding pack - framing report, data dictionary, decision log and model card - so a new joiner reaches context in days rather than weeks.

5.5 Working with marketing and customer support

Department	What they give the project	What they get back	Ritual
Marketing / CRM	Campaign calendar, incentive budget, historical response data, holdout discipline	Weekly scored list with bands and reason codes, campaign uplift analysis	Weekly campaign sync + sprint demo
Customer support	Ticket data, qualitative churn reasons, feedback on false positives	Score and top reason codes in the agent console, retention gestures within policy	Monthly feedback session, agent survey
Product management	Roadmap context, product telemetry	Ranked churn drivers that are product problems rather than marketing problems	Quarterly churn-driver review
Finance	Margin per category, cost of incentives, validation of the value model	ROI of retention spend, incremental revenue estimates	Quarterly value review
Legal / DPO	Consent framework, legal basis, DPIA input	Documented data usage, retention rules and activation constraints	Gate reviews G1 and G4

The feedback loop that matters most

Support agents are the only people who hear the real reason a customer is leaving. A simple structured field in the console - 'was this risk flag plausible?' - turns thousands of daily contacts into a qualitative validation set that no offline metric can replace.

6. Project governance and communication

6.1 Stakeholder map

Stakeholder	Interest	Influence	What they need	Channel and frequency
Executive sponsor (CMO / COO)	High	High	Does the investment pay back, and what is the risk	Monthly steering committee, one-page dashboard
Chief Data Officer	High	High	Is the strategic data programme delivering reusable foundations	Monthly steering + data governance board
Marketing executives	High	High	Usable segments, credible uplift, control over the offer	Weekly campaign sync, sprint demo
Retention / CRM team	High	Medium	A reliable weekly list they can action, with reasons	Weekly operational sync
Product managers	Medium	Medium	Which product issues drive churn	Quarterly churn-driver review
Data team	High	Medium	Clear definitions, sustainable pipelines, no shadow data	Daily stand-up, sprint rituals
Technology / infrastructure	Medium	High	Security, scalability, integration with the existing stack	Architecture review at G2 and G4
Customer support	Medium	Low	A tool that helps rather than another screen	Monthly feedback session
Finance	Medium	Medium	Cost control and a defensible ROI	Quarterly value review
Legal / DPO	Medium	High	Compliance evidence before go-live	Gate reviews, DPIA
Customers (indirect)	High	Low	Relevant contact, not harassment; transparency and the ability to object	Privacy notice, preference centre, contact frequency cap

6.2 Governance instances

Instance	Frequency	Participants	Mandate and decisions
Steering committee	Monthly	Sponsor, CDO, marketing exec, AI PM, PO, finance	Budget, scope, gate go / no-go, arbitration between departments
Project team rituals	Daily stand-up; 2-week sprint planning, review, retrospective	Squad + PO	Sprint scope, impediments, continuous improvement
Weekly campaign sync	Weekly	PO, retention manager, data analyst, data scientist	Which segments are actioned this week, holdout size, feedback on last week
Data governance board	Monthly (existing body)	CDO, data stewards, DPO, data engineering	Approves new data sources, definitions, catalogue entries, quality KPIs
Architecture review	At gates G2 and G4, then on major change	Technology and infrastructure team, MLOps, security	Approves the target architecture, security and scalability

Instance	Frequency	Participants	Mandate and decisions
Model validation committee	Before each production promotion	Lead data scientist, PO, DPO, an independent reviewer	Reviews the model card, metrics per slice, leakage checklist and limitations; approves or refuses promotion
Privacy / ethics review	Before go-live, then annually	DPO, legal, PO, lead data scientist	DPIA, legal basis, fairness, customer transparency, contact policy
Run review	Monthly once live	MLOps, data scientist, retention manager	Incidents, drift, retraining decisions, cost

Why a separate model validation committee

The people who build a model are the worst-placed to notice that it is too good to be true. A short, standing review with an independent reader - armed with the model card and the leakage checklist - is the cheapest defence against deploying a model that has learned the future.

6.3 Communication plan

Audience	Message	Format	Frequency	Owner
Steering committee	Value delivered, risks, decisions needed	One-page dashboard: churn rate, uplift vs. control, ROI, RAG status	Monthly	AI PM
Marketing and retention	Who to contact, why, and what happened last time	Scored list in the CRM + campaign results pack	Weekly	PO / analyst
Support agents	How to read a risk band and its reasons	Console tooltip, one-page cheat sheet, short training	At release + on change	PO / UX
Data and technology teams	Technical state of the system	MLflow UI, model card, Sphinx documentation, architecture decision records	Continuous	Lead data scientist / MLOps
Product and finance	Churn drivers and economics	Quarterly review deck	Quarterly	PO / analyst
Whole company	What the project is and what it is not	Short internal note at go-live	Once, then at major milestones	AI PM

6.4 Communicating model outputs to technical and non-technical audiences

	Technical audience	Non-technical audience
Unit delivered	Calibrated probability, SHAP values, full metric set	Risk band (high / medium / low) and rank, never a raw probability
Explanation	Global and local SHAP, feature importance, partial dependence	Top three reason codes in plain language: 'fewer visits than usual', 'two late deliveries', 'stopped buying in their main category'
Performance	PR-AUC, lift curves, calibration plots, slice tables	'Of the 10,000 customers we contact, about 2.5 times more are genuine churners than if we picked at random'
Uncertainty	Confidence intervals, stability analysis	'This is a probability, not a prediction - some flagged customers will stay, and that is expected'

	Technical audience	Non-technical audience
Channel	MLflow UI, model card, repository documentation	CRM fields, BI dashboard, monthly one-pager, agent console
Caveat that must always travel with the output	Correlation is not causation; the model ranks risk, it does not prescribe treatment	'The model tells us who is at risk, not what to offer - the offer policy is ours'

7. AI project management methodology

7.1 Chosen methodology

The project uses **CRISP-DM as the lifecycle**, executed with **Scrum for the build phase** and **Kanban for the run phase**, with five explicit phase gates. CRISP-DM supplies the phases and their feedback loops, which pure Scrum does not model; Scrum supplies the cadence, the demo and the stakeholder rhythm, which CRISP-DM does not provide; Kanban fits the interrupt-driven nature of monitoring, incidents and retraining once the model is live.

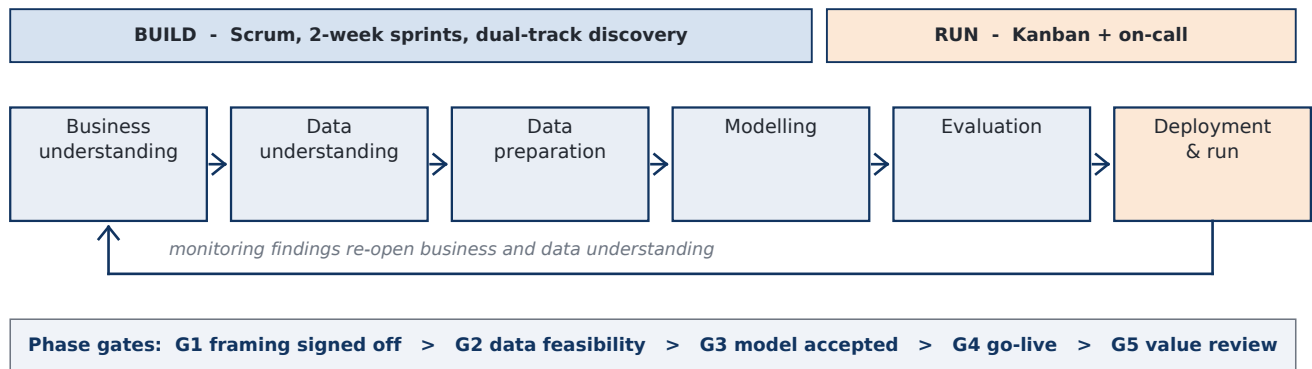


Figure 2 - CRISP-DM lifecycle executed with Scrum during build and Kanban during run, with five gates that release budget rather than merely mark progress.

Gate	Question answered	Exit criteria
G1 - Framing	Are we solving the right problem	Churn rule agreed, KPIs and baseline measured, value case quantified, DPIA started
G2 - Data feasibility	Does the data support the question	Sources accessible, identity resolution match rate acceptable, point-in-time dataset built, quality checks in place
G3 - Model accepted	Is the model good enough and safe enough	Beats the rule baseline, passes the leakage checklist and slice floors, model card written, committee approval
G4 - Go-live	Can we operate it	Pipelines automated, monitoring live, runbook and rollback tested, privacy sign-off, campaign and holdout designed
G5 - Value review	Did it work	Measured uplift against control, ROI, decision to scale, re-frame or stop

7.2 Why this methodology suits an AI project

Property of AI projects	Consequence	How the methodology answers it
The outcome is uncertain until the data is seen	A fixed scope and a fixed date cannot both be promised	Short sprints with a time-boxed exploration budget; gates that allow an early stop
Data understanding changes the problem	Requirements are discovered, not collected	CRISP-DM's explicit loop back from evaluation to business understanding
Model quality is not the deliverable; the business change is	A model can be accurate and worthless	Business KPIs reviewed in the same meeting as model KPIs; G5 measures uplift, not accuracy
Deployment and monitoring are where most projects fail	'Done' cannot mean 'the notebook works'	Deployment and run are first-class CRISP-DM phases with their own gate and their own Kanban

Property of AI projects	Consequence	How the methodology answers it
The work is cross-functional	Handovers destroy context	One squad, one backlog, shared definition of done
Run work is unpredictable	Sprints break when incidents arrive	Kanban with WIP limits and a service class for incidents after go-live

Alternatives considered. A pure waterfall plan was rejected: it would force the churn definition and the model choice to be fixed before any data has been examined. Pure Scrum without CRISP-DM was rejected because it gives no place to data understanding and treats deployment as just another story. A scaled framework such as SAFe is disproportionate for a single squad. The Google MLOps maturity model is used as a **target**, not as a process: the project aims at level 1 (automated pipeline) for v1, with level 2 (automated CI/CD for models) as the v2 objective.

7.3 Risk register

#	Risk	P	I	Mitigation	Owner
R1	Target leakage produces an excellent offline model with no real skill	M	H	Point-in-time snapshots, leakage checklist at G3, independent review, scepticism about unusually high scores	Lead DS
R2	Data access or integration is slower than planned	H	H	Data access requested during framing, phased source integration, v1 trained on transactions and behaviour only	Data Eng.
R3	Model is accurate but never used by marketing	M	H	Retention manager embedded from day one, scores delivered inside the CRM, adoption tracked as a KPI	PO
R4	Low precision wastes incentive budget	M	M	Threshold chosen on expected value, banded actions, cheap nudges for medium risk, uplift modelling in phase 2	CRM manager
R5	Privacy or compliance breach	L	H	DPIA before production, legal basis documented, consent enforced at activation, no special-category data	DPO
R6	Model bias against a segment (country, tenure, channel)	M	M	Slice evaluation with floors, fairness review at G3 and monthly	Lead DS
R7	Drift after peak season degrades performance silently	H	M	Drift monitoring, year-on-year comparison, scheduled and triggered retraining	MLOps
R8	Scope creep towards uplift, real-time and deep learning	H	M	Written non-goals, roadmap v1 / v2 / v3, PO arbitration at sprint planning	AI PM
R9	Infrastructure or compute cost overrun	M	M	Cost per run tracked as an MLflow tag, monthly cost review, batch before real-time	MLOps
R10	Key person dependency	M	M	Pair work, code review, documentation generated automatically, no notebook-only knowledge	AI PM
R11	Holdout contaminated or abandoned under commercial pressure	M	H	Holdout written into the campaign process and the definition of done; steering committee informed if it is removed	PO
R12	The model changes the behaviour it predicts (self-fulfilling effect)	M	M	Control group used as the reference population for evaluation and label construction	Lead DS
R13	External data source breaks or licence lapses	M	L	External features optional by design, pipeline degrades gracefully	Data Eng.

#	Risk	P	I	Mitigation	Owner
R14	Stakeholders over-trust the score	M	M	Caveats travel with every output, bands instead of raw probabilities, training for agents	AI PM

P = probability, I = impact (L / M / H). The register is reviewed at every sprint review and formally at each gate.

7.4 Handling cost and planning derivation caused by model iterations

Model iteration is where AI projects lose control of their budget: there is always one more feature to try and one more hyper-parameter to tune, and no natural stopping point. The counter-measures are contractual rather than technical.

Mechanism	How it works in practice
A 'good enough' bar agreed in advance	At G1 the business commits to the threshold that makes the project worth running - lift @10% of at least 2.5. Once it is reached, effort moves from tuning to deployment and adoption
Time-boxed spikes	Exploration is a story with a fixed budget (for example five days) and a written outcome. Extension requires an explicit decision, not a silent continuation
Phase gates that release budget	Each gate releases the budget for the next phase only. A project that fails G2 costs the framing, not the whole plan
Diminishing-returns rule	If two consecutive iterations improve the primary metric by less than a defined margin, modelling is frozen and the effort is redirected to data quality or activation, which usually pay more
Compute cost tracked per experiment	Cost recorded as an MLflow tag alongside metrics, so the cost of a marginal metric gain is visible rather than invisible
Scope laddering	v1 batch, tabular, one model; v2 real-time, survival, uplift; v3 sequence models. Ideas are not refused, they are scheduled
Planning buffer and re-forecast	20% contingency on the data phases, re-forecast at each gate with actuals, communicated to the steering committee as a range rather than a single date
Reuse over rebuild	Feature pipelines, registry and monitoring are built once for the programme (objective O5), so their cost is amortised over later use cases

7.5 Trello board - simulation of the project

A Kanban board simulates the management of the churn project from framing to run. The structure below mirrors the CRISP-DM phases and the gates; the board is the working artefact, this table is its specification.

Lists (columns)

List	Purpose	WIP limit
Backlog	Everything identified but not yet scheduled, ordered by value	-
Sprint backlog	Committed for the current 2-week sprint	Sprint capacity
In progress	Actively being worked on	4
Review / validation	Code review, data quality validation, model validation committee	3
Blocked	Waiting on access, a decision or an external dependency, with the blocker named	-

List	Purpose	WIP limit
Done (sprint)	Meets the definition of done	-
Run / monitoring	Live operational work: drift alerts, retraining, incidents	3

Labels, custom fields and conventions

- **Labels** - Data, Modelling, MLOps, Business, Governance / Privacy, Risk, Spike.
- **Custom fields** - CRISP-DM phase, gate, sprint, owner, effort (story points), objective (O1-O5).
- **Card template** - user story, acceptance criteria, definition-of-done checklist, links to the MLflow run and the pull request.
- **Gate cards** - each gate is a card in Review / validation with the exit criteria as a checklist; it cannot be moved to Done until every item is ticked.
- **Risk cards** - the register of 7.3 is mirrored on the board so that mitigations are actual work items with owners.

Representative cards by phase

CRISP-DM phase	Cards on the board
Business understanding	Measure the current 90-day churn baseline Agree the churn definition with marketing and finance Quantify the value case Define KPIs and acceptance thresholds Map stakeholders and the communication plan Gate G1
Data understanding	Inventory sources against the catalogue Request data access Build the customer ID graph and measure the match rate Profile quality on orders, clickstream and support Assess external sources Gate G2
Data preparation	Build the curated order and behaviour tables Implement point-in-time snapshots Feature family: RFM and trends Feature family: engagement decay Feature family: service friction Automated data quality expectations Leakage checklist
Modelling	Rule-based baseline Logistic regression baseline LightGBM v1 with MLflow tracking Hyper-parameter spike (5 days, time-boxed) Calibration SHAP explainability and reason codes Slice and fairness evaluation
Evaluation	Out-of-time evaluation Expected-value threshold selection with marketing Model card Model validation committee Gate G3
Deployment	Package the project (cookiecutter, dependencies, PEP8) Register the model in the MLflow registry Weekly batch scoring job CRM integration of scores, bands and reason codes REST serving endpoint for the support console Shadow run for one cycle Runbook and rollback test Privacy sign-off Gate G4
Run and monitoring	Data quality and drift dashboards Alert routing and on-call First campaign with randomised holdout Uplift analysis Monthly retraining automation Champion / challenger cycle Gate G5 value review

Board link

The Trello board built for this case study is public and available at <https://trello.com/b/134SCaGh> - 51 cards across the seven lists, with the CRISP-DM phase, objective, effort and sprint recorded as custom fields. It is deliberately shown mid-project (sprint 5, modelling phase): gates G1 and G2 are closed with their exit criteria ticked, G3 to G5 are still open, and two cards sit in Blocked with the blocker named.

8. Conclusion

The framing presented here treats customer churn prediction at RetailGenius as an operational change supported by a model, rather than as a modelling exercise. Its defining choices are a churn definition agreed with the business rather than inferred by the data team, a metric set that keeps business value visible next to model performance, a batch-first architecture that reuses the existing ETL and governance assets, a monitoring plan that accounts for a 90-day label delay, and a measurement discipline - the randomised holdout - that makes the value of the project falsifiable.

The same choices are what make the project a credible first step for the wider strategic data programme: the feature pipelines, the registry, the monitoring and the governance rituals built here are the foundations that the next use cases - recommendation, fraud detection, demand forecasting - will reuse. Part 2 of this graded project implements this design as a production-oriented Python project with versioning, a standard structure, reproducible environments, generated documentation, and MLflow tracking, registry and serving.

9. References

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Governance, privacy and documentation

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- Regulation (EU) 2024/1689 (**AI Act**) - risk classification; commercial churn prediction is not listed as high-risk, but transparency and governance practices are adopted voluntarily.
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Tools

- Trello - trello.com (Kanban board for the project simulation).
- MLflow, scikit-learn, LightGBM, SHAP, Sphinx, black / flake8 / pylint, Poetry - used in Part 2.
- Cookiecutter Data Science - drivendata.github.io/cookiecutter-data-science/

Use of generative AI

Generative AI (Claude, Anthropic) was used to help structure this report, draft and reformulate sections, and challenge the completeness of the framing. All analytical choices - the churn definition, the KPI set, the architecture, the methodology and the risk register - were reviewed, corrected and validated by the group, and the sources above were checked independently.